INTRODUCTION TO INFORMATION GEOMETRY

Log-Odds and Exponential Updating

Week 5 · Lecture 2

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Odds and log-odds

For a Bernoulli law $(x, 1-x)$, the odds of A against B are

$$\frac{x}{1-x}.$$

The log-odds coordinate is

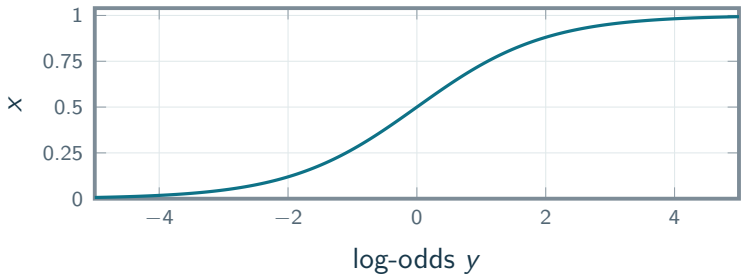
$$y = \ln \frac{x}{1-x}, \qquad 0 < x < 1.$$

As x ranges through the interval, y ranges through $\mathbb{R}$.

From the log-odds line to the interval

Solving for x gives

$$x = \frac{e^y}{1 + e^y}.$$



A particular log-odds value

Find the log-odds for a probability of $x = 0.2$:

$$y = \ln \frac{0.2}{0.8} = ?.$$

Is this state above or below the equal mixture $x = \frac{1}{2}$?

A particular log-odds value

Find the log-odds for a probability of $x = 0.2$:

$$y = \ln \frac{1}{4} = -\ln 4.$$

It is negative, corresponding to $x < \frac{1}{2}$.

The replicator equation in log-odds

Differentiate $y = \ln x - \ln(1-x)$:

$$\frac{dy}{dx} = \frac{1}{x(1-x)}.$$

Since $x' = x(1-x)(f_A - f_B)$, the chain rule gives

$$y' = f_A - f_B.$$

The log-odds rate is exactly the relative fitness advantage.

Constant advantage becomes linear motion

If $f_A - f_B = k$ is constant, then

$$y' = k, \qquad y(t) = y_0 + kt.$$

Converting back to probability gives

$$x(t) = \frac{e^{y_0 + kt}}{1 + e^{y_0 + kt}}.$$

This is the logistic solution from Week 1.

Exponential updating for many types

With constant fitness values $f_1, \dots, f_n$,

$$p_i(t) = \frac{p_i(0) e^{f_i t}}{Z(t)}.$$

The normalizing factor is

$$Z(t) = \sum_j p_j(0) e^{f_j t}.$$

The exponential weights remain positive and sum to one after normalization.

An exponential update with three types

Let $P(0) = (.2, .3, .5)$, $f = (1, 0, -1)$, and $t = \ln 2$. Compute the unnormalized weights and the updated law:

$$(p_i(0)e^{f_i t})_i = (\ ?,\ ?,\ ?), \qquad P(t) = ?.$$

An exponential update with three types

Let $P(0) = (.2, .3, .5)$, $f = (1, 0, -1)$, and $t = \ln 2$. The weights are $(.4, .3, .25)$, with $Z = .95$.

$$P(t) = (\tfrac{8}{19}, \tfrac{6}{19}, \tfrac{5}{19}).$$

The log-normalizer describes mean fitness

Differentiate $Z(t)$:

$$\frac{d}{dt} \ln Z(t) = \sum_i p_i(t) f_i = \bar{f}(t).$$

Differentiating again gives

$$\frac{d^2}{dt^2} \ln Z(t) = \sum_i p_i(t) (f_i - \bar{f}(t))^2.$$

The second derivative is the fitness variance.

Derivatives of a log-normalizer

At $t = 0$, let $P = (.2, .3, .5)$ and $f = (1, 0, -1)$. Compute $\bar{f}$ and $\mathrm{Var}_P(f)$.

$$\bar{f} = ?, \qquad \mathrm{Var}_P(f) = ?.$$

Derivatives of a log-normalizer

At $t = 0$, let $P = (.2, .3, .5)$ and $f = (1, 0, -1)$.

$$\bar{f} = .2 - .5 = -.3.$$

$$\mathrm{Var}_P(f) = .7 - (-.3)^2 = .61.$$

A mean constraint produces exponential weights

To maximize entropy subject to a fixed expectation of values z_i, use a Lagrange multiplier λ. The resulting probabilities have the form

$$p_i = \frac{e^{\lambda z_i}}{\sum_j e^{\lambda z_j}}.$$

The multiplier is chosen so the required mean is attained.

Entropy need not move monotonically

For a two-type constant-advantage system with $k > 0$, $x(t)$ increases toward 1. Starting from $x(0) = .2$, the binary entropy first increases until $x = \frac{1}{2}$, then decreases. Entropy concerns the current distribution of outcomes. The replicator field determines the direction of its change.

The transition to relative entropy

The log-probability $\ln p_i$ appears in surprisal, entropy, log-odds, and exponential updating. In Week 6, we compare two entire probability laws using expected differences of their

log-probabilities.